



TO: The Executive Board, SmartBridge AI Core
FROM: Techbridge University College, ICT Division
DATE: 7 May 2026
REFERENCE: TUC-ICT-PROP-2026-002
SUBJECT: Sovereign Deployment Framework for Ghana's One Million Coders Programme

1. STRATEGIC VISION

Techbridge University College proposes a structured alliance with SmartBridge AI Core to serve as the unified technical and operational partner for the One Million Coders Programme in Ghana. This proposal sets out the framework, delivery model, and projected outcomes of a collaboration designed to transform the initiative from a generic skills-delivery exercise into a nationally-aligned programme of industrial and digital capacity building.

The Government of Ghana's ambition is not simply to produce coders — it is to cultivate a generation of sovereign industrial architects capable of building, owning, and exporting technology rooted in Ghanaian industrial context. Achieving this requires a partner with established local infrastructure, validated delivery experience, and deep alignment with national policy. Techbridge provides each of these, having already trained over 100 students under the CTVET programme and operating an active cohort of 15,000 enrolled across our Oyibi campus.

This alliance moves beyond the standard SaaS procurement model. By combining SmartBridge's AI-native learning platform with Techbridge's institutional infrastructure and regional reach, the partnership creates a delivery model that is simultaneously world-class in technical capability and sovereign in its operation.

2. THE 6R DELIVERY FRAMEWORK

We propose the 6R Strategic Delivery Framework as the operational structure for the alliance. Each stage represents a distinct phase of programme development, with clear ownership and measurable outcomes:

- **Review — Demand Mapping:** A structured audit of over 1,000 local SMEs and government agencies to identify the precise AI-native and vocational competencies required across Ghana's growth sectors, ensuring the programme responds to genuine market demand.
- **Reduce — Curriculum Pruning:** Removal of legacy theoretical content and proprietary software dependencies from existing curricula, replaced with AI-augmented, open-source pathways that reduce the barrier to technical mastery by an estimated 45 per cent.
- **Refine — Vertical Specialisation:** Integration of Techbridge's CAD/CAM and Digital Twin capabilities into specialised coding tracks aligned with the Jewellery, Fashion, and Product Design sectors — Ghana's highest-value industrial export categories.
- **Reuse — Node Activation:** Deployment of Techbridge's eight regional campus clusters as Sovereign Access Nodes, providing high-bandwidth infrastructure and GPU rendering capacity throughout Greater Accra, Kumasi, and the Northern Region.
- **Regenerate — IP Genesis:** Establishment of joint Sovereign Tooling Laboratories in which students design and build the enterprise software and industrial tools used within their own sectors — creating a domestic IP ownership tier.

- **Reinforce — Policy Alignment:** Integration of real-time Ministry monitoring dashboards to enable automated certification alignment with Ghana's national vocational standards framework, providing government with direct programme oversight.

3. PROGRAMME DELIVERY AND OPERATIONAL SYNERGY

The alliance will operate under a Sovereign Execution Model structured around four core principles:

- **Joint Curriculum Engineering:** SmartBridge's global algorithmic learning engine will be merged with Techbridge's industrial design frameworks, producing a curriculum that is technically rigorous and contextually relevant to Ghana's economic priorities. Content development will be jointly owned and locally hosted.
- **Physical Infrastructure Utilisation:** Techbridge's campuses will serve as hands-on industrial laboratories, providing equipment-based learning that complements the digital delivery platform. This is particularly critical for the Jewellery Design, Fashion Technology, and Product Engineering tracks, where physical tools and materials cannot be replicated digitally.
- **Data Sovereignty and Compliance:** All student records, learning pathways, and assessment data for the programme's one million participants will be stored and managed within Ghanaian jurisdiction, in full compliance with the Ghana Data Protection Act. Techbridge will serve as the designated Local Data Steward under the alliance agreement.
- **National Policy Synchronisation:** Every programme KPI will be aligned with the Ministry of Trade's strategy for sectoral export development and the Ministry of Education's digital literacy targets, ensuring government accountability and a clear pathway to programme sustainability beyond the initial funding period.

4. THE FOUR SPECIALIST DEGREE PROGRAMMES

Techbridge contributes four active, accredited degree programmes to the alliance, each engineered for Ghana's emerging digital sectors. These programmes are currently operational at the Oyibi campus and provide the industrial backbone for the specialised coding tracks:

- **Product Design and Entrepreneurship:** Integrating UI/UX design methodology with AI-native product development workflows and Ghanaian market commercialisation strategies. Graduates are equipped to launch and scale digital products within the domestic SME sector.
- **Fashion Design Technology:** Combining digital textile engineering, 3D garment visualisation, and smart supply chain technology into a creative technologist's curriculum. West Africa's fashion sector is projected to generate 120,000 new artisanal roles by 2030; this programme prepares graduates to lead that transition.
- **Jewellery Design Technology:** Applying high-precision CAD/CAM modelling and generative design tools to Ghana's gold and artisan sector. The programme creates a structured pathway from traditional craft to export-grade industrial production, supporting an estimated \$450 million export potential.
- **Digital Media and Communications Design:** Equipping graduates with AI-driven storytelling, generative video production, and brand architecture skills for the West African media and advertising market. The programme draws on Techbridge's existing partnerships with Ghanaian broadcasters and digital agencies.

Each programme is structured as a four-year degree with built-in industry placements and live project briefs drawn from Ghanaian enterprises. All four are eligible for direct alignment with the One Million Coders Programme certification framework.

5. ECONOMIC IMPACT AND PROJECTIONS

The alliance model creates measurable economic value at three levels — national, sectoral, and operational — each of which is supported by independently modelled projections:

NATIONAL DIGITAL GVA <i>\$0.8B → \$7.0B</i> Projected growth in Ghana's digital Gross Value Added from 2024 to 2030, positioning Ghana as the continental benchmark for technical-vocational sovereignty.	IMPORT SUBSTITUTION <i>42% reduction</i> Projected reduction in foreign software procurement dependency over five years, generating an estimated saving of \$800 million for the Ghanaian economy.
LOCAL ECONOMIC RETENTION <i>60% of fees</i> Proportion of programme service fees retained within Ghana's domestic economy through local hiring, infrastructure investment, and tax contributions — versus near-total capital flight under direct international procurement.	ALLIANCE COST EFFICIENCY <i>GHS 455M vs GHS 840M</i> Estimated 45 per cent cost reduction against equivalent direct international procurement, whilst maintaining full technical capability and programme scope.

The full Economic Impact Model, including year-by-year GVA trajectory, sectoral contribution breakdown, and workforce absorption projections, is available in the accompanying digital presentation at the reference above.

6. CONCLUSION AND CALL TO ACTION

Techbridge University College is prepared to begin deployment immediately. Our existing infrastructure, trained personnel, and active student cohort provide a platform that requires no build phase — only integration with the SmartBridge delivery engine. The full programme can be operational within eight weeks of alliance ratification.

We propose an immediate technical synchronisation workshop to finalise the Sovereign Access Node deployment plan and initiate the SME Demand Mapping audit. Following workshop agreement, joint curriculum engineering can commence within the first fortnight.

This is not a capacity-building exercise in the conventional sense. It is a structural transformation of Ghana's technical workforce — with sovereignty, accountability, and long-term export capability built into the architecture from the outset. The distinction between a domestic alliance and a direct international procurement is the distinction between a programme that builds Ghana and one that merely operates within it.

We invite the SmartBridge Executive Board to ratify this alliance and authorise commencement of Phase One.

Authorised by:

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In co-ordination with the Ministry of Trade and Industry, Ghana

